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MARCH 2018

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A National Machine Intelligence Strategy for the United States

AUTHORS

William A. Carter
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William Crumpler
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A REPORT OF THE
CSIS TECHNOLOGY
POLICY PROGRAM

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Executive Summary

The United States is at the precipice of a defining moment in history.

Over the past five years, progress in machine intelligence (MI) has greatly accelerated. From the defeat of Go champion Lee Sedol by DeepMind's AlphaGo program to the first deployments of fully autonomous vehicles on public roads, recent events are challenging us to reevaluate what may soon be possible for computerized systems. MI systems have already begun to quietly pervade a growing share of businesses, governments, and individual lives around the world, and we are only just beginning to grasp the impacts that this technological revolution will have on our economy, our society, and our national security.

The United States' current economic and technical lead in MI uniquely positions us to ensure the technology's impact will be broadly beneficial, for Americans and the world. Other countries have already established national strategies to capture the benefits of MI and catch up with the United States. Capturing this opportunity requires we invest in remaining at the leading edge of MI's development, and plan for the broader changes it implies. A national strategy for MI will provide the United States a platform for creating the policies and investments that ensure MI's progress is consistent with American goals, norms, and values.

IV

This strategy should have two overarching goals. The first is to promote the safe and responsible development of MI technology by funding long-term R&D where the private sector is not incentivized to invest, developing a workforce for the MI age, creating dynamic commercial markets for MI technologies to capture the innovation of the private sector, and proactively managing the risks and disruptions that MI will bring. The second goal is to maintain U.S. leadership in MI by reinforcing our innovation base and establishing strategic partnerships to leverage the comparative advantages of our allies and lead the development of global MI governance.

A. INVEST IN THE CONTINUING RESEARCH AND DEVELOPMENT OF MI TECHNOLOGY.

Though the United States benefits from strong private-sector investment in MI R&D, companies alone cannot carry the burden of keeping the United States at the forefront of MI innovation. The government has a unique role to play in funding research into defense-specific innovations, systems of MI ethics and control, and high-risk, high-reward long-term research with uncertain commercial returns. The federal government must not waver in its commitment to advancing these essential domains, and must ensure that our national security community is able to keep up in the development and deployment of MI-enabled systems.

B. DEVELOP A WORKFORCE FOR THE MI AGE.

We must adapt to shifting economic circumstances to prepare our workforce for the jobs of the future. This will require a renewed emphasis on computer science education and technical skills to build and maintain MI systems, as well as basic digital literacy and the liberal arts as the economy faces a growing demand for adaptable workers with soft skills capable of complementing the operations of MI systems. It will also be essential for policymakers to help workers displaced by automation to develop new skills by expanding retraining and continuing education initiatives and strengthening the social safety net. We must also ensure that our country is able to attract and retain the best talent from around the world.

C. CREATE A FLEXIBLE AND OPEN DATA ECOSYSTEM IN WHICH MI CAN THRIVE.

An essential component for the development of MI systems is access to large amounts of high-quality training data. By expanding government open data initiatives, working with the private sector to support private-private sharing in ways that protect privacy, and supporting the development and deployment of new standards to improve data quality, the government can help MI developers access the data they need to create new and more powerful MI applications.

D. CREATE THOUGHTFUL PUBLIC POLICY TO SPUR MI ADOPTION AND REDUCE BARRIERS.

Many U.S. enterprises are held back from capturing the full benefits of MI due to uncertainty over legal and regulatory requirements and a lack of supporting IT infrastructure. To help grow the market for U.S. MI developers and improve productivity throughout the economy, policymakers should consult with industry partners to identify ways to remove legal and regulatory uncertainty, and devise ways of supporting IT modernization within the private sector.

E. DEVELOP PROACTIVE STRATEGIES TO MANAGE MI'S RISKS.

While the apocalyptic warnings voiced by some in the tech community are overblown, MI systems will raise new challenges in the areas of privacy, algorithmic bias, system safety and control. The U.S. government can help confront these risks by leading in the development of safety, ethics, and control standards for MI, and working with the private sector to develop methods of testing and certification for MI systems. In addition, policymakers must establish clear expectations for MI developers in the event of accidents in order to reduce regulatory uncertainty and manage public anxiety.

F. LEAD THE DEVELOPMENT OF MI TECHNOLOGY AND MI GOVERNANCE AROUND THE WORLD THROUGH STRATEGIC MI PARTNERSHIPS.

By developing strategic MI partnerships, the United States can leverage the comparative advantages of our allies and help shape global technical and governance standards around the development and use of MI. A United States with a clear national strategy for MI should play a central role in the development of MI technology and MI policy around the world, and that will require strong partnerships with our allies.

► **IT** will take time to craft and implement all of the elements of a national MI policy, but we must act now to establish a cohesive national MI strategy. As a first step, the president should designate an MI lead at the White House to coordinate the development and implementation of a strategy and ensure that this is a priority across the federal government. This person should prioritize reviewing and implementing the 2016 AI R&D Strategy recommended by the National Science and Technology Council. Federal agencies and regulators should also initiate consultations with the private sector and MI experts in academia to ensure that our approach is flexible enough to adapt over time and appropriate to the technology. These discussions will take time, but will help to guide the efforts of policymakers and ensure that MI governance develops in partnership with industry, not in opposition to it. ▲

SECTION I

Engingeer
working on
machine in
factory.



VI

INTRODUCTION

The Promise and Peril of Machine Intelligence

1

MI refers to machines' ability to perform tasks that would normally require human intelligence. Computer scientists and mathematicians develop MI systems by imparting the ability to find patterns in large data sets to computers ("machine learning"). Although the underlying techniques used to create MI have existed since the 1960s, improvements in computing power, availability of data, and machine learning have spurred an MI renaissance over the past five years, generating the first applications of MI viable outside academic labs.

MI's advance has been rapid. Machines can already support or replace human tasks and decisionmaking, making us more incisive, productive, and free from rote work. In the future, MI will scale further, pairing human-like reasoning with growing volumes of data from sensors embedded in devices, vehicles, robotics, and infrastructure. With increasing ubiquity, MI will gather, interpret, and act on data from the physical world with minimal human

intervention. We will allow machines to make many decisions because they are faster at and more capable of interpreting a world quantified by massive volumes of data.

These changes are remarkable in both their promise and their danger.

► **INTELLIGENT** machines' efficiency and scale will help us address some of the world's most urgent problems. With reams of data from x-rays and MRIs, cancer researchers are teaching MI to detect the earliest signs of disease in patients. Agricultural companies are using millions of photos of plants to teach autonomous drones what healthy plants look like, improving farming practices and increasing crop yields. Energy experts are using data from building sensors and roads to create machines that can silently monitor and improve our infrastructure's electricity economy. Across sectors, new applications of MI that will meaningfully improve human life are reported almost every day.

"Artificial intelligence is the future not only of Russia but of all of mankind."

RUSSIAN PRESIDENT **VLADIMIR PUTIN**,
SEPTEMBER 2017

2

But MI's impending ubiquity also carries risks. When MI makes mistakes, produces unintended consequences, or is used nefariously, it will do so on a grand scale. MI systems' efficiency will leave little time for human intervention; its complexity will make it difficult to understand and remediate failures after the fact. Ill-advised applications of MI and technical mistakes have already disrupted financial markets, eroded public trust, and threatened our national security (see previous page). Even with advanced planning, problems with the technology cannot be fully avoided. In the next decade, we will likely see MI eliminate certain jobs, manipulate our personal information, limit our insight into how organizations make decisions, and reduce equity in parts of society. If not properly controlled, the technology

holds the potential to exacerbate existing economic polarization, disenfranchisement, and social fragmentation.

How policymakers prepare for MI's impact today will determine whether the technology's benefits ultimately outweigh its risks. The United States' allies and competitors recognize this, and are vying to shape the norms and values that dictate MI's use. China announced its national artificial intelligence (AI) strategy in the summer of 2017, highlighting how it could apply the technology to track people on closed-circuit cameras, censor the internet, and predict crimes. In the fall of 2017, Russian president Vladimir Putin emphasized the strategic importance of MI in a speech to Russian schoolchildren. Canada, France, and the United Kingdom have each announced national-level strategies for enhancing their positions in the technology's development and ensuring its use preserves their norms around privacy, equity, and transparency.

► **TO** manage the social, geopolitical, and economic outcomes of MI for Americans and the world, the United States must also create a vision and strategy for the future of intelligent machines. We must invest in the research and development (R&D) of MI and the establishment of a technological ecosystem in which MI can thrive. We must renew our commitment to education and to the development of a workforce that possesses not only the technical skills to build intelligent machines, but the creativity and ingenuity to apply them to unique problems. We must take the lead in driving the development of national and international standards and policies for governing and managing risk and unintended consequences. Our action now will dictate MI's impact on the world for decades to come. ▲

Financial Markets

Unexpected interactions between superfast trading algorithms have been implicated in multiple instances of erratic activity in global stock markets of the past five years, including May 6, 2010 “Flash Crash,” which saw the U.S. stock market plunge 998.5 points (roughly 9%, or USD \$1 trillion in market value) in minutes.

Left: Traders work on the floor of the New York Stock Exchange before the closing bell May 6, 2010.

PHOTO BY MARIO TAMA/GETTY IMAGES

<https://www.sec.gov/news/studies/2010/marketevents-report.pdf>



3

Public Trust

Both applications of MI and the mechanisms by which it is created have disrupted the public’s confidence in institutions. In 2017, the U.K.’s National Health Service (NHS) and Google’s MI unit, DeepMind, were censured by the British Information Commission (ICO) for inappropriately using 1.6 million patients’ records to train MI algorithms for a health application, spurring outcry among patient advocates.

Left: Royal Free hospital in north London said it would change how it handles data in future after ICO’s findings.

PHOTO BY BEN STANSTALL/AFP/GETTY IMAGES

<http://www.bbc.com/news/technology-40483202>



National Security

Political twitter bots have been used in attempts to influence the outcome of elections around the world across the past five years, including in Turkey, Mexico, India, and the United States. Few of these influence campaigns have harnessed rapidly advancing machine learning or MI technologies, but there is concern that they soon will, making it more difficult to distinguish real accounts from sophisticated MI systems that learn from and mimic human behavior.

Left: A committee hearing on ‘Russia Investigative Task Force: Special Media Companies’ held on Nov. 1, 2017.

PHOTO BY ALEX WONG/GETTY IMAGES

<https://arxiv.org/pdf/1706.05143.pdf>





THE STATE OF MACHINE INTELLIGENCE

4

▲
Alan Turing,
a British
mathematician,
logician and
cryptographer.

A Brief History of MI

The study of MI dates to at least 1950, when famed British computer scientist Alan Turing considered the question “Can machines think?” in his paper *Computing Machinery and Intelligence*. A few years later, in 1956, Dartmouth University hosted the first academic conference on MI, signaling its start as a formal area of research. Since then, researchers in universities, government agencies, and technology companies have worked to realize real-world applications of MI. Several problems, however, consistently impeded the technology’s progress.

MI relies on a technique called machine learning, in which algorithms parse large data sets to identify and learn from patterns. Until recently, the available body of digital data from which machines could learn was limited. Even in those instances in which data were available, computer processors ran too slowly to make most MI applications feasible. It could take days or even weeks for a computer to churn through the data needed to learn a single MI task.

MI Today

Since 2010, a proliferation of data from connected devices and improvements in computing chips have created the circumstances MI needed to thrive. These developments allowed for the introduction of a new machine learning technique, deep learning, which has been integral to MI's increasing ubiquity. Deep learning algorithms are generally more accurate and precise than other forms of machine learning, suiting them to commercial use.

With the necessary puzzle pieces in place, MI has sprung from relative obscurity onto the international stage. The number of MI papers published annually has increased more than nine-fold since 1996. In the last few years alone, MI has met or exceeded human capability in a range of narrow tasks, including object detection and speech recognition.

MI's ascendance is not limited to academia. Investors have poured more than \$5 billion into start-ups developing commercial MI applications over the past five years. Microsoft's head of research, Peter Lee, compared the cost of hiring a top MI expert to that of signing an NFL quarterback. American companies acquired more than 50 early-stage MI companies between 2012 and 2017, many of which had not yet earned any revenue, the focus resting instead on gaining access to MI ideas and talent. By 2021, MI software, hardware, and services are expected to command nearly \$58 billion across the global economy.

► **AMID** the feverish excitement that has come to surround MI, there have been many exaggerated claims about its capabilities. It is important to recognize that, while transformative, machines' form of intelligence is

The Complementary Nature of Human and Machine Intelligence

HUMAN INTELLIGENCE

🧠 Derives from observation, evolution, culture, and more

🧠 Operates well in novel, chaotic, and unpredictable environments

🧠 Attuned to perform work which requires judgment, creativity, empathy, interaction, and multi-domain expertise

🧠 Transferable and generalizable across multiple domains

MACHINE INTELLIGENCE

🤖 Derives from data and algorithms

🤖 Operates well in predictable environments richly documented in historical data

🤖 Attuned to perform work which requires precision and efficiency surrounding a narrow task

🤖 Restricted to the narrow problem space it was created to address (i.e., an algorithm created to translate language cannot also identify objects in images)

"MI will radically transform society, but it will do so by complementing, not replacing us."

7

much different from—and much more rudimentary than—that of humans. MI will radically transform society, but it will do so by complementing, not replacing us.

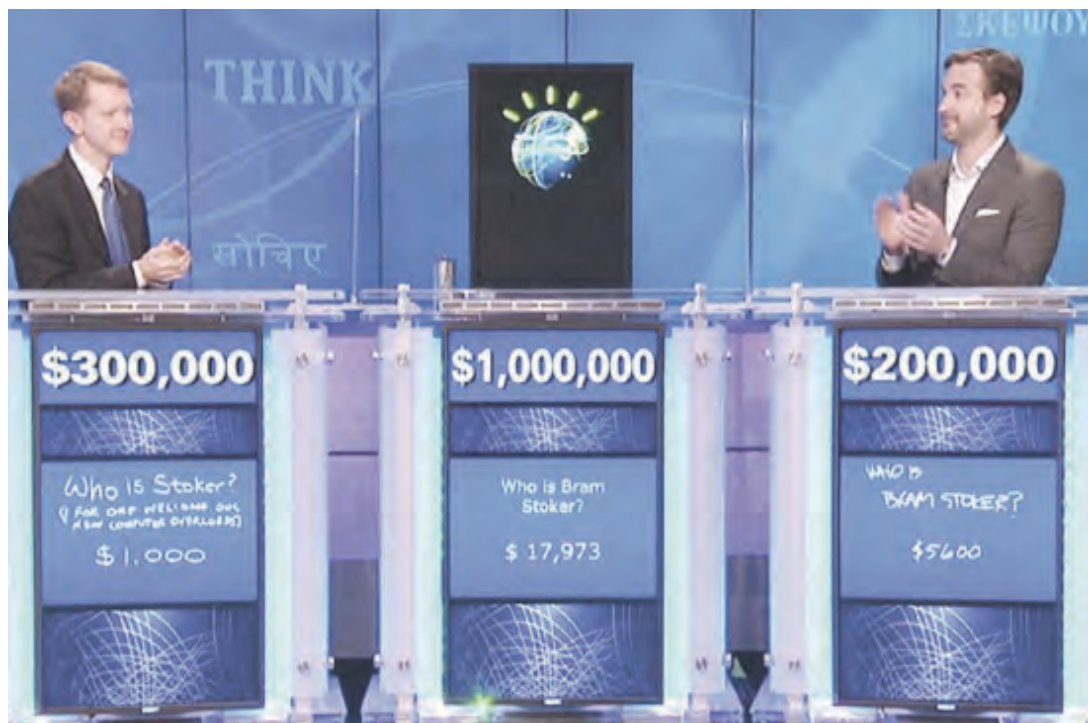
What MI Can Do

Today's MI systems are exceptionally good at pattern recognition, which suits them to two categories of work.

The first category of work comprises rote and repeatable tasks. MI's ability to automate pattern-matching tasks, like recognizing objects in images and transcribing speech, will eliminate many tedious aspects of our jobs. MI-enabled transcription, for example, can help doctors save time recording patients' medical files. Machine precision at these forms of repetitive work complements human capabilities. Unlike humans, machines do not become fatigued, bored, or distracted when performing dull tasks. When we delegate such work to machines, we can devote more attention to the higher-or-

der work machines can't perform. With the aid of automated systems to review luggage scans for suspicious objects, Transportation Safety Authority (TSA) agents, for example, may spend less time staring at x-ray screens and focus more time on interacting with passengers and assessing possible threats.

The second category of pattern-matching work MI can perform is the real-time analysis of massive volumes of data, which is impossible for humans today. For example, Amazon uses MI to monitor and analyze many thousands of competitors' online storefronts in real time, allowing them to dynamically change product prices in response to market conditions. Cyber security companies like Cylance monitor organizations' network activity to identify and preempt threats as they develop. MI's ability to perform this new category of work will enable the delivery of new kinds of products and services that make people and organizations more efficient, dynamic, and productive.



What MI Cannot Do (Yet)

While MI's accomplishments are impressive, these advances should not inspire fears about sentient robots supplanting humans.

Individual MI algorithms generally perform a single function. An MI algorithm designed to identify a roadside object, like a stop sign, could not also identify an open space in a parking lot. MI systems comprise a series of these algorithms, each working in relative isolation. As a result, machines are not able to piece together context in the way humans can.

- 8 Take, for example, the IBM Watson computer's 2011 performance in a game of Jeopardy. The MI system dominated the competition against two human Jeopardy champions until the final Jeopardy question: "Its largest airport is named for a World War II hero; its second largest, for a World War II battle." Watson guessed the answer to be Toronto, Canada. Unfortunately, the category was "U.S. Cities." The system got the answer wrong because it was unable to integrate the broader context

of the category into its narrow "input-response" paradigm.

MI systems are also limited by their reliance on massive volumes of training data. While a human child could be taught to recognize the meaning of a stop sign given a few examples, a machine might require thousands of photos to achieve the same feat. As a result, MI is not well-suited to tasks for which there is little historical data.

► **FINALLY,** it is worth emphasizing that MI systems are not sentient: they lack self-awareness, motivations, and intentions. They cannot explain their actions or mistakes. They are entirely reliant upon the algorithms we program into them and the data we feed them. Fears that robots will come to resent and overtake humans are unwarranted. Without very substantial, nonlinear advancements in the technology, the MI of science fiction will remain fictional. ▲

▲
IBM's
Watson
competing
in a
game of
Jeopardy.
**PHOTO BY
VINCENTLTLE**

"Tasks for [MI] systems are often framed in narrow contexts. . . . While machines may exhibit stellar performance on a certain task, performance may degrade dramatically if the task is modified even slightly. For example, a human who can read Chinese characters would likely understand Chinese speech, know something about Chinese culture and even make good recommendations at Chinese restaurants. In contrast, very different AI systems would be needed for each of these tasks."



IMPACT ON THE ECONOMY, SOCIETY, AND NATIONAL SECURITY

10

▲
An experimental Ford Fusion self-driving delivery car is displayed at CES in Las Vegas, Nevada, Jan. 12, 2018.

PHOTO BY
DAVID MCNEW/AFP/
GETTY IMAGES

MI Will Transform the Economy in Two Waves

11

The first wave of economic impact from MI is already occurring and will expand rapidly in the next 5 to 10 years as organizations invest heavily to develop or buy applications of the technology. Companies around the world are pouring money into MI R&D amid growing competition to be first-to-market with MI products and services. The technology sector, including giants like Google and Baidu, spent close to \$30 billion on creating and deploying MI in 2016 alone. Other industries are investing heavily, as well. Toyota and Ford each spent over \$1 billion to develop autonomous vehicles in 2017.⁷ Last year, global revenues from the sale of early MI products and services reached over \$12.5 billion, and venture capital and private equity investment in the technology surpassed \$6 billion. Nvidia, which dominates global market share for the computer chips that power MI, saw its stock performance increase by over 700 percent from 2014 to 2017.

The next wave of economic impact will occur as organizations move from developing MI and selling narrow, MI-enabled consumer products (e.g., Amazon's Alexa and autonomous drones) to implementing MI as an enterprise system. Today, less than one-quarter of companies have adopted MI, and most organizations that are using it are doing so in narrow-use cases within IT departments, like in cyber security. Widespread MI adoption will have a transformative impact that parallels the adoption of the public internet, which went from relatively narrow use in the early 1990s to dictating much of global business operations by the first decade of the twenty-first century.

► **THERE** is debate among economists about the nature of the impact this wider-spread MI adoption will have on economic growth and jobs.

Optimists argue a virtuous cycle will emerge in which MI makes organizations more productive by automating work, which will in turn reduce the cost of goods and services, increase real income, and enhance people's average standard of living. Some have posited that this economic boost could increase global GDP by as much as 14 percent (\$15.7 trillion) by 2030. They argue that those individuals whose jobs are automated will find new jobs in new categories of work that do not yet exist, and that the technology will not increase total unemployment. The average worker in 2015, for example, could en-

joy the same annual income level as a worker in 1915, while working just one-third of the year. As jobs in agriculture and manufacturing were destroyed by technology in 1930s, the service sector exploded, growing by nearly 600 percent in the last 80 years.

Pessimists take a dimmer view, arguing that MI could upend the economy by sewing widespread unemployment. A widely shared 2013 report from Oxford economists suggested 47 percent of American jobs could be automated through MI and related technologies in

"The nature of these changes, positive or negative, will depend in large part on the actions taken by policymakers and governments today."

the next 20 years. Experts in this camp believe MI is fundamentally different from previous technological breakthroughs in the speed and scale with which it will affect jobs. They argue society's most vulnerable workers, who perform low-wage, repetitive jobs, will be hit hardest, putting millions out of work while enriching a small set of capital owners. They point out that even as technological marvels have abounded, real wages for most workers have barely increased since the mid-1960s, and annual U.S. productivity growth decreased from 3 percent between the end of World War II and 1973 to half that level today.

The most extreme scenarios envisioned by both the optimists and the pessimists are unlikely to come to pass, but we can expect MI to significantly change work, jobs, and the economy. The nature of these changes, positive or negative, will depend in large part on actions taken by policymakers and governments today. Investment in new education and training programs and expanded support programs for workers in transition, for example, could limit exposure to MI-driven unemployment. The expansion of multidisciplinary degree programs in computer science and the social sciences at public universities could spur informed students to start new companies and create new job opportunities. Proactive policy solutions can reduce the risk of MI's painful disruptions as MI is implemented across the economy.

MI Will Transform Our Lives in Positive Ways, but Also Comes with Risks

MI's impact will extend beyond the economy, making possible advances that will transform our everyday lives. For consumers, MI promises greater choice and more convenient and personalized goods and services. Today, these benefits manifest as recommendations in our Netflix queues or Amazon shopping carts, but in the future more meaningful applications will become common.

One application of MI that could have far-reaching consequences for our quality of life is enhancements to the health care we receive. Doctors and researchers are developing MI systems that could improve the detection of disease, find new drug treatments, and create personalized therapies to match patients' unique symptoms and genetic attributes. These benefits could extend to mental health care as well. Psychologists are using MI virtual therapists to help troops discuss their post-traumatic stress disorder (PTSD) when they are too uncomfortable to do so with people. Facebook has launched an MI system to detect suicidal posts from users before they're reported. Doctors are using natural language processing (NLP) MI to detect patterns of speech indicative of depression in patients.

MI could also provide new solutions to global challenges like climate change. For example, shared fleets of autonomous vehicles could reduce the number of cars on the road, limiting congestion and air pollution and creating the opportunity to turn parking lots into green spaces. Scientists are experimenting

**SECTION
III**



14

▲
Pilot models of the Uber self-driving car is displayed at the Uber Advanced Technologies Center on Sep. 13, 2016 in Pittsburgh, Pennsylvania.

**PHOTO BY
ANGELO
MERENDINO/AFP/
GETTY IMAGES**

with using MI to detect and monitor invasive species in our oceans, protecting fisheries and tourism areas relied on by local economies. Energy experts are developing MI systems to enable efficient smart grid technology, reducing energy consumption and limiting the acceleration of climate change. Companies like Microsoft are making massive investments in MI to help people around the world monitor, model, and manage the

Earth's natural systems, decreasing the damaging impacts of natural disasters, which cost \$306 billion in 2017.

These are just a few of the many ways MI will improve our daily lives. Academics, researchers, and corporations are finding ways to use MI to improve cybersecurity, financial services, education, transportation, infrastructure, manufacturing, and many other fields. But we

must acknowledge that the benefits MI offers us are accompanied by tradeoffs.

First, MI's reliance on data creates challenges around privacy and equity. Amid the rush to create MI applications, organizations could use our personal information in ways that surprise and upset us. Google's DeepMind MI unit and the U.K.'s National Health System (NHS), for example, were recently cited for using the sensitive health records of more than 1.6 million patients without notifying the patients affected. MI's reliance on data will also affect citizens' rights to equitable treatment. Many MI systems rely on data produced by humans—data that reflects our biases and errors in judgment. When biased data is used to train machines, these biases can reach massive scale. Experts worry, for example, about the increasing use of MI to make hiring decisions could replicate and scale patterns of discriminatory hiring.

► **SECOND,** MI systems will complicate our ability to assign responsibility for decisions and to receive transparency into how decisions are made. Who, for example, will be responsible when an autonomous vehicle crashes? Should responsibility fall with the original equipment manufacturer, software maker, or the passenger who failed to push a software update? When assigning accountability to a machine or its user, it will often be difficult or even impossible to determine how the MI arrived at a decision. MI systems are complex, sometimes opaque even to their creators. In domains ranging from medical diagnostics to loan risk scoring, this lack of transparency will create significant safety and legal risks as failures become inexplicable and encoded biases become calcified within audit-resistant models.

Third, MI's impact on our work will spur existential questions. For millennia, human beings have largely defined our value and purpose based on our labor and our unique ability to reason and solve problems. All modern economies are organized around the principle of maximizing the productivity of human beings and distributing the benefits of human-driven production. While current MI technology is limited and brittle, it is already eroding some forms of work, and as the technology evolves to tackle more complex and dynamic problems, it could challenge how we relate to our jobs and organize society.

Finally, as with all technological developments, some will use MI for nefarious purposes. Automation is already being employed in cyber-attacks, helping hackers attack large numbers of targets quickly and cheaply. MI systems can also create near-perfect counterfeits of audio and video, which could exacerbate growing public distrust in news media and information shared by organization.

MI's effect on privacy, equity, accountability, transparency, and public trust will spur significant concern and debate in the public and private spheres in the coming decade. People will look to their elected officials for leadership and guidance in managing the effects of MI on their lives at work and at home.



MI Will Redefine the Balance of Both Economic and Military Power Around the World

MI is creating a new era of strategic technological competition that, like the space race for the previous generation, will define the balance of global power for decades. Policymakers around the world are taking notice. In September 2017, Russian president Vladimir Putin said, “Whoever becomes the leader in [MI] will become the ruler of the world.”

China’s New Generation of Artificial Intelligence Development Plan, released in 2017, echoes this view, saying, “Machine intelligence [is] the strategic technology that will lead in the future.”

Nations’ ability to produce and leverage MI technologies will define their economic competitiveness. Going forward, the global distribution of wealth, income, and soft power will be increasingly determined by nations’ ability to develop innovative MI applications and leverage those applications across their economies. In the United States, the tech sector directly accounts for 7.5 percent

of GDP and provides over 7 million jobs, and is a key driver of growth and competitiveness across virtually every other sector. In sectors as diverse as manufacturing, health care, energy, retail, and advertising and media, deployment of MI technologies holds enormous promise for driving new growth.

Eight of the top 10 tech companies in the world are U.S. based, which allows the United States to exercise disproportionate power in cyberspace. U.S. companies reach virtually every consumer in the world, helping us to spread our values around the world. Global supply chains enabled by technology and MI have driven global economic integration. Information and communication technology (ICT) infrastructure and digital inclusion are also key components of our efforts to build influence and support development and human rights in the developing world.

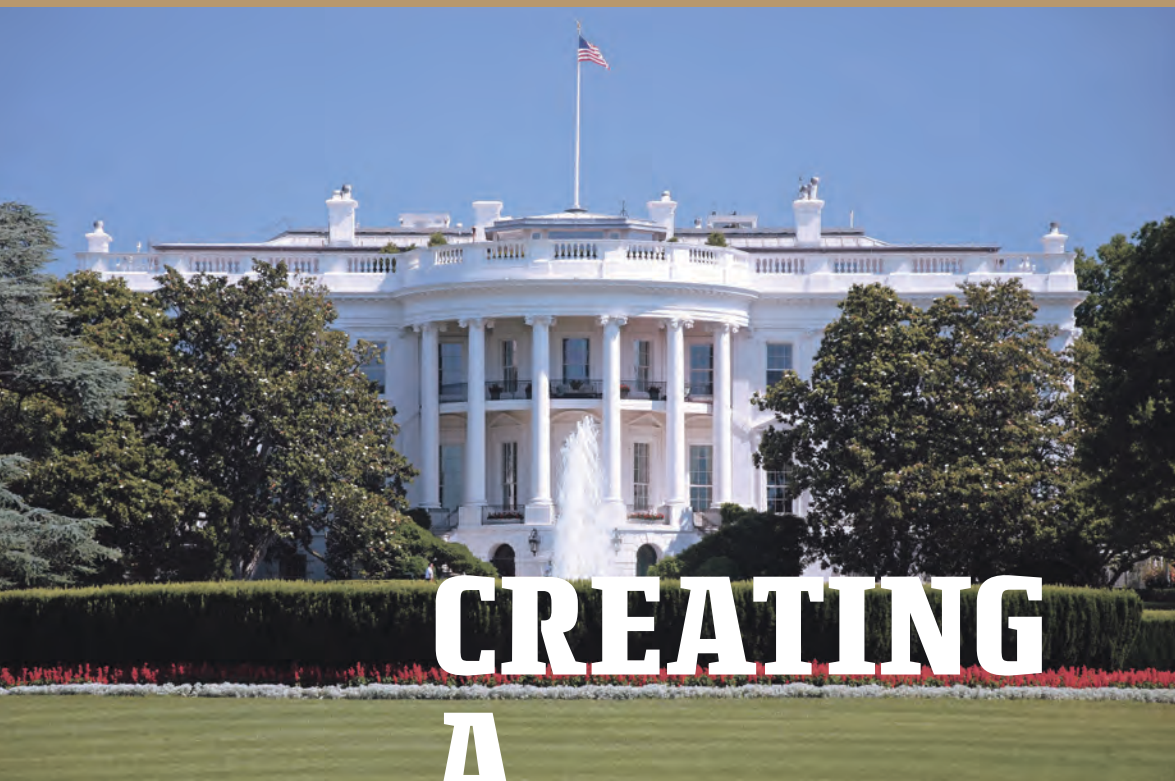
MI also has significant implications for hard power. Across the world, from the Chinese People's Liberation Army (PLA) to the U.S. Department of Defense (DOD), there is growing recognition that the next generation of military technologies will be driven by MI. The nature of conflict is shifting from today's "informatized warfare" to "intelligentized warfare" or "algorithmic warfare." Drone swarms, autonomous supply convoys, and remote medical care technologies have all seen significant investment from the world's major militaries.

These changes will have profound impacts on the nature of conflict. Not only will military platforms and weapons change, but MI systems will fuel the development of entirely novel strategies, tactics, and concepts of operations. Perhaps the most transformative applications of military MI are in command and

control (C2). MI-enabled C2 could develop entirely novel strategies, anticipate enemy tactics, accelerate intelligence, surveillance, and reconnaissance (ISR), and help coordinate activities of large numbers of dispersed units acting in tandem. As a greater share of decision-making on the battlefield happens at machine speed, human thinkers may be unable to keep up.

► **THE** focus of the policy debate around military applications of MI has centered around lethal autonomous weapons systems (LAWS). On one hand, many worry about the long-term implications of designing machines to kill human beings, citing the risks of losing control of the technology and of malicious actors being able to wreak havoc and create carnage at scale. On the other hand, some see an opportunity to free human soldiers from the most dangerous combat situations, help to alleviate the risks of human bias and misperception in combat, and for nations like Russia and China, an opportunity to upend the traditional military dominance of the United States by creating weapons that can fight and adapt at machine speed.

There is also growing concern MI will also empower nonstate actors to develop new asymmetric warfare tactics. Proliferation of MI capabilities in the commercial sector will provide broad, low-cost access to intelligent systems. Just as cyber capabilities have enabled small groups to damage national security on a disproportionate scale, MI may spur new and unexpected threats. MI's capacity to create indiscernible counterfeits of audio and video provide one example. In 2017, Adobe demonstrated a new product that, with 10 minutes of audio, can replicate a person's voice exactly in limitless artificial audio. ▲



CREATING A NATIONAL STRATEGY

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To shape MI's impact on our economy, national security, and society, the United States requires a comprehensive national strategy to coordinate efforts across the federal government and the private sector. This strategy should have two overarching goals. The first is to promote the safe and responsible development of MI technology. MI promises tremendous benefits, and capturing these should be a top priority for the U.S. government. The second is to maintain U.S. leadership in MI. MI will transform both economic and strategic power around the world, and leadership in MI technology will be essential to securing our future.

Promoting the safe and responsible development of MI technology will require us to reinforce our innovation base by funding long-term fundamental R&D in areas where the private sector is not clearly incentivized to invest. We must develop a workforce with the skills to both build and maintain innovative MI systems and work with and complement them in jobs that are not easily automated. We must create a flexible and open data ecosystem in which data-intensive machine learning algorithms can thrive, while protecting privacy and preventing discrimination. We must also support the development of dynamic commercial markets for MI technologies

Map of National Strategies

Highlighted countries have released AI strategies



to leverage the innovative capacity of the private sector while ensuring that the technology is designed with security and safety in mind. Finally, we must anticipate and proactively manage the inevitable risks of MI technologies. MI will cause accidents, disruptions, and unintended consequences, and if we do not have a strategy to deal with these challenges in advance, we are likely to fall into the trap of reactionary policymaking that poses the greatest risk to the development of the technology.

Maintaining U.S. leadership will require a forward-thinking, proactive approach to MI. The United States' dominant position in the global tech industry did not come about by accident. U.S. tech innovators benefit from strong investment in research and development, a flex-

ible regulatory environment that allows companies to innovate and adapt quickly to new technologies, and perhaps most importantly, a highly educated, dynamic workforce. Our national strategy should build on and reinforce those strengths. We must also establish strategic MI partnerships to leverage the comparative advantages of our allies, promote our values around MI development, and help shape global technical and governance standards around the development and use of MI. A United States with a clear national strategy for MI should play a central role in the development of MI technology and MI public policy around the world, and that will require strong strategic partnerships with our allies.

QUICK GLANCE

The United States needs a comprehensive national strategy for machine intelligence to achieve two key goals:

1. Promote the safe and responsible development of MI technology.
2. Maintain our global leadership in MI.

A.

Invest in the Continuing Research and Development of MI Technology

STRATEGY
A

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The seemingly sudden explosion of MI applications over the past five years was not the result of a single technological breakthrough. Instead, it represents the culmination of technological trends that created an environment in which MI could thrive. A national strategy must not only maintain but also actively build upon this technological environment.

The first step for policymakers is to develop a whole-of-government strategy for investment in fundamental MI R&D. Countries including Singapore, Japan, the United Kingdom, Canada, France, and Korea have all announced major MI funding initiatives in the past two

years, representing well over a billion dollars of backing for new R&D projects. China dwarfs all other countries in the scale of its research ambition. After the July 2017 announcement of the country's new MI development plan, local and provincial governments announced billions of dollars of support to the industry, with the cities of Xiangtan and Tianjin alone pledging a collective \$7 billion to MI projects.⁷ In comparison, total U.S. government R&D investment in MI was \$1.1 billion in 2015, and the Trump administration's proposed budget would have cut the National Science Foundation's MI research funding by 10 percent. The future of MI will be heavily shaped

by those at the forefront of its development, and we risk falling behind by underinvesting at this critical time.

Invest in government MI R&D that complements private-sector R&D.

Corporate research labs have filled part of the gap in MI R&D, and venture capitalists are investing heavily to bring new MI applications to market. Since 2012, venture capitalists have spent nearly \$6 billion backing start-ups creating commercial MI applications. The largest companies in the technology sector, including Facebook, Apple, Google, Amazon, and Microsoft, spent an estimated \$20–30 billion on MI in 2016.

But private-sector investment alone is not enough. The federal government has a unique role to play in supporting the development of MI technology,

and it must start by updating our research funding priorities to adapt to the unique challenges and opportunities of MI. The government is also singularly positioned to support research on the development of accountable, ethical or privacy-restrained MI that has little commercial advantage, but significant societal benefit.

Private investors focus primarily on projects with immediate and guaranteed commercial applications. There is little incentive for most companies to fund sustained work on exploratory projects with long incubation periods or uncertain prospects for returns. The federal government is in a unique position to support basic research that may not pay off for 20 to 30 years, but, like the internet, may prove revolutionary. As a 2016 National Science and Technology Council report on MI R&D reported, the economic benefits of MI research will more than justify its costs, even in the current environment of cost-cutting across government. According to some studies, cur-

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rent levels of R&D spending are just one-half to one-quarter of the level that would produce optimal economic growth.

The U.S. government should fund a new generation of fundamental research into MI through universities, national labs, and the National Science Foundation. Providing researchers with access to federal institutions' supercomputing resources could also help drive innovation in the field. The U.S. government can also support private-sector innovation through cooperative research and development agreements (CRADAs), loan guarantees and research grants from organizations like the Small Business Administration (SBA). The SBA's grant programs, including both the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs, are major sources for startup support, representing the largest source of nondilutive, early-stage seed capital in the world.

► **OUR** MI R&D strategy should also support the development of technologies and infrastructures that support MI development. The U.S. discussion around MI has focused heavily on algorithms, but algorithms are one part of broader technological ecosystem. Federal investment should support not just direct research into MI algorithms and the computing chips that power them (Graphics Processing Units, or GPUs) but also research in fields like materials science, batteries, high-performance and quantum computing, robotics, miniaturization, and sensors that complement MI.

R&D investment can also help to keep the U.S. government abreast of innovation in the private sector. The U.S. private sector is our greatest competitive edge in MI technology, and the U.S. government has long struggled to innovate internally. MI technologies will be essential for the U.S. government as it seeks to cut costs, generate new operational efficiencies for government agencies, and protect the nation and our citizens. Expanding funding to federal government programs like In-Q-Tel, Defense Innovation Unit-Experimental (DIUx), Defense Advanced Research Projects Agency (DAR-

PA), and Intelligence Advanced Research Projects Activity (IARPA) to invest in MI research and MI startups will not only help to fund private-sector innovation, but will also strengthen ties between the government and MI innovators, keeping the U.S. government abreast of developments in the private sector and helping the U.S. government to leverage these advances.

Ensure that DoD and the national security community do not fall behind in the development and deployment of MI-enabled systems.

The government also has a unique interest in MI for national security applications, and we must ensure that we do not fall behind the development of MI-enabled systems. This will require us to rethink our approach to bringing private-sector innovations into DoD. The Chinese government has recently taken steps to promote what they call "military-civil fusion" by creating opportunities for government and military researchers to partner with leading technology companies in the development of new products. China's approach is based off of similar efforts in the United States, notably through programs like DoD's DIUx and In-Q-Tel, which offer the security establishment a way to bypass cumbersome contract processes and accelerate the deployment of cutting-edge technologies within our military.

But these programs are tiny compared to the behemoth of traditional federal acquisitions, and we cannot expect a tiny part of the federal acquisitions process to produce the bulk of our next generation of military capabilities. As the share of transformative innovation continues to shift toward the commercial sector, these lean, agile acquisition programs cannot remain the exception, they must become the norm, and we must expect and tolerate stumbles and problems as we learn to leverage these approaches at scale.

We also cannot afford to adopt dogmatic views of MI technologies that prevent us from exploring their potential. In the United States, lethal autonomous weapons systems (LAWS) often dominate conversations around military MI, and discussions around MI policy too often revolve around the dangers of “killer robots.” This conversation is simultaneously premature and behind the times. Fully autonomous weapon systems remain far from combat-ready, and human soldiers are not going anywhere anytime soon. At the same time, the technology for private individuals to build simple autonomous killing machines at little cost already exists, as a professor from UC Berkeley ironically demonstrated while advocating against the development of autonomous weapons.

In 2016, the secretary of defense said, “whenever it comes to the application of force, there will never be true autonomy, because there’ll be humans (in the loop).” This argument makes sense for today’s AI technology. Human-machine partnerships are far more effective and reliable than fully autonomous systems in complex and dynamic

combat environments, and are likely to remain so for a while. But if we rule out the possibility of fully autonomous combat systems, DoD risks missing out on a class of technologies that could fundamentally transform warfare.

Other countries like Russia and China are unlikely to exercise the same restraint when it comes to fully autonomous weapons systems, which they view as an opportunity to leapfrog U.S. military dominance. If these countries were to field fully autonomous weapons systems that could analyze and adapt to our tactics and strategies at machine speed, it could render our defenses ineffective if we do not do the same. DoD should invest in the next generation of combat systems that leverage the full potential of AI, including the potential future development of lethal AI systems that can operate without humans in the loop. Instead of LAWS “never,” our policy should be “not until they can outperform human/MI collaboration,” including making ethically acceptable choices about when to “pull the trigger.” ▲

RECOMMENDATIONS

Invest in the Continuing Research and Development of MI Technology

1. Fund long-term fundamental research where the private sector is unlikely to support adequate investment.
2. Invest in technologies that support MI development.
3. Ensure U.S. government access to cutting-edge technologies developed by the private sector.
4. Ensure that DoD and the national security community do not fall behind in the development and deployment of MI-enabled systems.

B.

Develop a Workforce for the MI Age

STRATEGY
B

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Though most conversations about leadership and competitiveness in MI focus on the latest technological advances, what these breakthroughs really come down to are the people. It is the quality of our human talent that has enabled the United States to achieve such a dominant position in the field of MI, and it will be the quality of our workforce that will allow us to translate these MI technologies into growth and prosperity for our country.

As we plan for the workforce of the future, we need to think about both educating the next generation of workers and retraining the current generation. While predictions of a “jobspocalypse” are overblown, it is impossible to deny that the accelerating adoption of MI and automation will cause significant changes in the U.S. workforce. It is like-

ly that new technological advances will create more than enough jobs to make up for those lost through automation, but current workers will confront obstacles as they struggle to transition to new careers. Promoting investment in continuing education and job retraining will be crucial to managing these disruptions, and reforms to our country’s education system and social safety net may be required to ensure that all workers have the opportunity to quickly transition to new roles.

Educate the MI workforce of the future.

Too often the debate around MI and education focuses on science, technology, engineering, and math (STEM) degree



programs, but STEM is both too broad and too narrow a term. While there is a shortage of qualified computer scientists and MI experts, fields like the physical and life sciences are already oversaturated with graduates. Furthermore, the emphasis on hard skills and the sciences fails to address the need for workers that can complement MI technologies. Hard skills are often the easiest to replace with automation, while soft skills like empathy, communication, and critical thinking will only grow in importance as jobs working with MI systems become more common.

Our education strategy should focus on a two-pronged approach: develop a strong pool of technical talent that can build and operate the technology of the future, and build a broad workforce of quick-thinking, adaptable people with basic digital literacy and the soft skills to complement and work with machines.

The first step is to build the pipeline of MI talent by expanding degree programs in computer science and MI. The United States currently confronts a major shortage of talent in computer science, and machine intelligence in particular. There are more than 510,000

open computing jobs nationwide, but the number of computer science students graduating in 2016 had the capacity to fill less than 10 percent of that need. Even among computer scientists, direct experience and education in machine intelligence is rare. MI experts are in such short supply that some companies are offering up to \$500,000 in starting salary to attract new graduates in the field. While some progress has already been made in building out computer science degree programs and expanding the pipeline of future graduates, many current graduates leave school with only general skills, and require extensive on the job training to work in specialized fields like MI.

The federal government can support technology education by funding research on MI curricula design, and helping to accredit MI education programs. Policymakers should also address the rising issue of professor retention, which is threatening to hollow out educational institutions and cripple our nation's capacity to train successive generations of MI talent. Exploding salaries for MI experts at companies like Google and Facebook have lured a significant number of professors away from their universities. One striking example of this was Uber's 2015 recruitment of almost 30 percent of the staff at Carnegie Mellon's National Robotics Engineering Centre. To reverse this trend, policymakers must expand funding for university MI programs and research grants to ensure our nation does not lose its capacity to sustain its innovative edge.

We also need to invest in universal digital literacy from an early age. The MI-augmented jobs of the future will require at least a basic level of understanding of digital systems and MI. A recent study found only 30 percent of jobs

now require only low levels of digital skill, down from 56 percent in 2002, indicating a shrinking window of opportunity for students who emerge from school without the ability to effectively work with digital systems. This does not mean that everyone will need to be able to write MI code; far fewer jobs in the future will require programming a computer than will require interacting with one. But it does mean that students will need to have some exposure to programming to understand at a high level how digital systems work, what they can and cannot do, and how humans can effectively collaborate with them.

Training everyone to be a software developer would be counterproductive, but a basic understanding of computer science and MI cannot remain a niche interest or specialized skill, it must become part of the same universal basic toolkit as reading and arithmetic. From K-12 through undergraduate and graduate education, digital and MI literacy should be integrated into curriculum development and accreditation programs to ensure that the next generation, regardless of the field they go into, is able to participate in the MI-enabled economy of the future.

Besides basic digital literacy, many agree that the most important skills for future workers will be soft skills like creativity, critical thinking, emotional intelligence, and adaptability. These skills are difficult for machines to replicate but critical to most businesses, making them resilient to automation and increasingly valuable as the role of human workers shifts to complementing the capabilities of MI systems. The ideal profile of the worker of the future is someone with technical proficiency, rather than mastery, and a background that will allow them to be a lifelong learner capa-

ble of applying their creativity and critical-thinking skills to a constantly shifting set of jobs and tasks performed in partnership with machines. To support the growth of these skills, policymakers should reemphasize the liberal arts and multidisciplinary programs that concentrate on graduating adaptable students capable of thriving in an uncertain workforce environment.

In the long term, MI could transform education in two fundamental ways. First, MI systems could be integrated into the classroom, supporting human teachers and providing a personalized learning experience and curriculum suited to each individual student's needs. These MI teachers could also be integrated into online and virtual classes to reach a larger pool of students around the world. Second, as MI becomes ubiquitous in

the workplace, the accelerating pace of technological change will require continuous education and training throughout our careers to keep up with the technology and maintain our skills. The current education system, in which we are educated and build our core skillset in our early years and then work through the rest of our lives, may no longer be sustainable. The Department of Education should organize a working group to study the long-term implications of MI for the education system, and how policymakers can work with

"a basic understanding of computer science and MI cannot remain a niche interest or specialized skill, it must become part of the same universal basic toolkit as reading and arithmetic."

educators and industry to build an education system that both leverages the potential of MI in teaching and meets the needs of the MI economy.

Attract and retain the world's best MI talent.

We should also ensure that graduates of these programs stay in the United States to work, invest, and build businesses. One of the greatest strengths of the United States is our ability to attract talented and skilled workers from abroad. The United States is the number one immigration destination in the world, particularly for people with advanced education and skills, and we educate and train many of the world's leading entrepreneurs and innovators.

► **WHILE** countries like China are working to incentivize Chinese-born and Chinese-American tech entrepreneurs to return to China, the United States is increasingly pushing away foreign talent, including limiting H1-B visas. Efforts to create more opportunities for Americans to get high-paying jobs by limiting access to those positions for foreigners makes it difficult for U.S. companies to access elite MI talent from other countries, or forces them to invest in innovation centers in other countries that are more welcoming to foreign talent. One of the biggest beneficiaries is Canada, where companies like Google, Facebook, Microsoft, and Uber have set up innovation centers to get around U.S. visa restrictions.

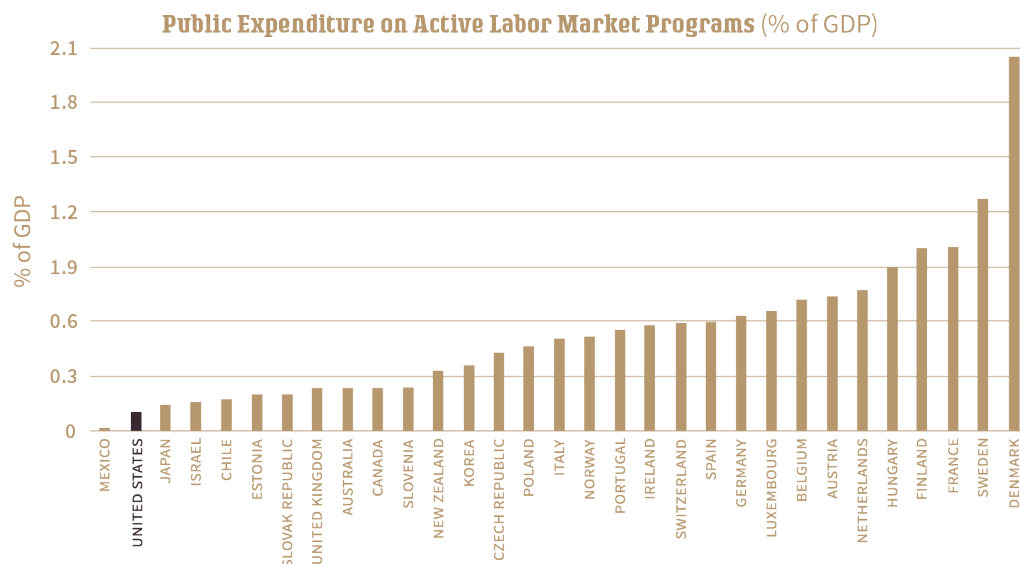
Cracking down on immigration will not only fail to create more opportunities for Americans to get high-paying jobs, it will stifle economic growth and competitiveness across the economy. In-

stead of trying to spur growth by turning away foreign workers, we need to embrace immigration and ensure that the next generation of MI entrepreneurs, wherever they are born, build the technology of the future in the United States.

One key step is to expand H1-B visas and make it easier for H1-B recipients to settle their lives and families in the United States. This will allow U.S. companies access to foreign workers while investing and creating jobs in the United States, instead of building



innovation centers and funding R&D overseas. We should also establish graduation visas to ensure that students who are trained and educated in our elite universities can put those skills to work in this country after they complete their degrees. Finally, we should expand incentives for entrepreneurs to set up businesses in the United States, including offering entrepreneurship visas to innovators who want to build businesses here.



NOTE: Data for Estonia and New Zealand from 2014; Data for UK from 2011

SOURCE: OECD Statistics (2015)

Reskill today's workforce for tomorrow's economy.

Surveys of company executives have consistently shown that lack of talent and technical knowledge is among the largest barriers to more widespread adoption of MI tools. In PwC's Digital IQ survey, only 20 percent of executives said their organizations had the skills necessary to succeed with AI. With the right talent, MI could transform most industries, allowing them to improve efficiency, provide more tailored, personalized products and services, improve quality control, and better manage their supply chains. Current workers may lack technical skills and digital literacy, but they understand the businesses, products, and customers that they work with, and that experience and exposure could be highly valuable to their companies if they can develop new skills.

We should also expand and reform federal retraining programs to best serve the needs of displaced workers. In its December 2016 report on artificial intelligence, automation, and the economy, the White House pointed out that the United States spends far less than other Organization for Economic Cooperation and Development (OECD) countries on active labor market policies designed to help workers navigate job transitions. OECD countries spent an average of 0.6 percent of GDP on such programs in 2014, compared to just 0.1 percent in the United States. Furthermore, while the pace of technological change has accelerated since the 1980s, federal retraining programs have stagnated. Relative to the overall economy, the United States spends less than half on such programs than we did 30 years ago.

Not all retraining programs are created equal, however. Recent assessments of the Workforce Investment Act (WIA) adult and dislocated worker programs have shown that the initiatives in question are generally ineffective at increasing worker earnings or employment rates. Part of the failure of these programs can be attributed to their organization as a system of discreet programs servicing different pools of workers. General savings and incentive programs for all workers to pursue continuing education and training throughout their careers could be more effective, and allow workers to identify the skills and expertise most relevant to the needs of their industries. The government should also improve job-search assis-

The government should work with private partners to develop guidance and provide resources for firms interested in developing talent within their own workforces to adopt MI tools. One example of this is the SBA's Small Business Technology Coalition, a partnership with companies like Google, Microsoft, and Facebook offering digital training resources to small businesses. Expanding and deepening the training provided through these types of programs could help many firms acquire the skills necessary to begin taking advantage of MI. The U.S. government should also work with the private sector to develop incentives like tax credits for firms that invest in continuing education and training to retool their workforces.

tance programs and other counseling resources helping match workers to jobs and reduce the time displaced individuals spend unemployed.

Community-based retraining initiatives with deep connections to local industry have been shown to be the most effective model for retraining. In addition to broad national-level programs to support the development of new skills, the U.S. government should establish programs to support community-based and state and local retraining efforts. The future of adult retraining will be led by community colleges developing partnerships with local employers to provide students with the skills they need to succeed in the jobs their communities require. These colleges will also be at the forefront of a new wave of online education, giving support to students turning to the emerging ecosystem of online education to re-skill and take advantage of new opportunities within their industry.

To support workers displaced by automation, policymakers should also consider expanding and reforming the social safety net to provide for the basic needs of citizens as they transition jobs or undergo retraining. Essential components of reform include unemployment insurance, work-sharing programs, welfare, and wage insurance. Strengthening unemployment benefits and welfare programs would help individuals be more resilient to job loss, allowing them the opportunity to retrain and seek new jobs without falling into poverty. These programs would also help insulate local economies from shocks caused by mass layoffs. Other measures like work-sharing programs could allow businesses to avoid laying off workers by instead reducing hours. Finally, initiatives like wage insurance could soften the blow for workers taking new, lower-paying jobs, and encourage displaced individuals to quickly reintegrate into the economy even if it meant taking a job where they would earn less. ▲

RECOMMENDATIONS

Develop a workforce for the MI age

1. Expand degree programs in computer science and especially machine intelligence.
2. Invest in universal basic digital literacy for the next generation.
3. Reemphasize the development of soft skills and the importance of liberal arts education.
4. Convene a DoE working group to study the long-term implications of MI for the nation's education system.
5. Create an immigration system that welcomes foreign talent.
6. Incentivize firms to invest in continuing education for their current workers.
7. Expand continuing education and retraining programs for today's workers, and the workers of the future.
8. Strengthen the social safety net to support workers in transition.

C.

Create a Flexible and Open Data Ecosystem in Which MI Can Thrive

STRATEGY
C

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An essential input to the development of MI systems is access to large amounts of high-quality training data. MI systems are data hungry. They require massive datasets to identify unique patterns and insights. For example, experts estimate autonomous vehicles will need to process 40 terabytes of data a day to drive around city streets absent our guidance. This is the equivalent of roughly 20 billion single-spaced pages of typewritten information.

The amount of data generated in the world has exploded as a result of the growth of smart phones and connected devices, the so-called internet of things (IoT). Ninety percent of the data in the world today has been created in the last two years, and that growth rate continues to accelerate. From 2011 to 2017,

the share of Americans who owned smartphones more than doubled, from 35 percent to over 77 percent. Combined with the proliferation of other internet-enabled devices, from toasters to cars, roads, and refrigerators, this trend has provided vast troves of data upon which to train MI algorithms.

One of the primary reasons for U.S. dominance in the field of MI is our country's unique data ecosystem, which affords MI researchers a large user base, expansive access to public data sets, and a supportive regulatory regime. In such an environment, researchers and developers are able to take full advantage of the country's accelerating data production to create MI technologies with unparalleled power and intelligence. Policymakers should build upon



this competitive advantage to ensure that U.S. MI developers continue to have access to the most open and flexible data ecosystem in the world.

One thing policymakers can do is expand access to datasets owned and managed by the U.S. government. Federal agencies hold many unique datasets that are valuable to MI developers, and opening these datasets up to the public could have an enormous impact on the creation of new MI applications. For example, when the government made the National Oceanic and Atmospheric Administration's (NOAA) weather and climate data publicly accessible, they created an industry worth over \$32 billion a year. Unlocking other federal datasets like unclassified

satellite imagery, transportation data, and financial information would create new opportunities for MI innovation and allow for dramatic improvements in the delivery of public services. The federal government can also work with state and local governments to explore new ways to share their datasets and integrate them with federal data across shared platforms.

Making this data available in easy-to-use, machine-readable formats is also important. The government can begin by harmonizing its own standards for structuring and labeling data, and should work with the private sector to create standards to enable sharing between the government and industry.

Policymakers can also play a role in removing barriers to the sharing of private-sector data between firms. Regulators and agency officials can promulgate standards and publish guidance to help firms navigate concerns over the applicability of competition or privacy laws when sharing customer data. Public-private partnerships can be leveraged to provide frameworks to facilitate the sharing of data between organizations, and incentives can be devised to encourage firms to make their data available through open application program interfaces (APIs). Though it will be important to develop privacy and data protection regulations for this new domain to safeguard consumers, policymakers will have to ensure that rules are tailored to avoid stifling innovation.

Ensuring a supportive ecosystem in the United States will be particularly crucial given the natural advantage some countries like China have due to the scale of their data production. China will be home to 20 percent of the world's data by 2020, and 30 percent by 2030, giving their MI developers the opportunity to access enormous datasets to train their models.

But U.S. innovators also have advantages. China may have a huge consumer market but U.S. companies have larger global market share. Chinese companies dominate the domestic market, but have struggled to compete globally. Chi-

"While China may have 3 percent of the world's data, U.S. companies have a tremendous head start on cornering the other 70 percent of the global data market."

na has 1.4 billion people, but Facebook has more than 2 billion users worldwide. Gmail has more users than any Chinese email service, and Facebook Messenger and WhatsApp have more users than any Chinese messaging service. As a result, while China may have 30 percent of the world's data, U.S. companies have a tremendous head start on cornering the other 70 percent of the global data market. These users are also more diverse and more global than users of Chinese services like WeChat and QQ. China's data is almost entirely on Chinese consumers, whereas U.S. companies like Facebook and Google have users around the world. This is an important advantage.

But U.S. tech companies' position in foreign markets is under threat from policies that make it more difficult for U.S. companies to compete. Pri-

vacancy advocates in the United States are also pushing for greater protections for data. Even close allies like the European Union are developing policies like the General Data Protection Regulation (GDPR) that create challenges for U.S. firms. The U.S. government must develop a strategy to combat protectionism, data localization, and privacy policies that harm our global tech companies. Doing this effectively will require balancing legitimate concerns around privacy and consumer protection both in the United States and abroad with the need for an open, flexible data ecosystem that supports innovation and experimentation in AI. We should also use trade agreements and establish bilateral and multilateral partnerships to promote the free flow of data and support collaboration in R&D for emerging technologies. ▲

RECOMMENDATIONS

Create a flexible and open data ecosystem in which MI can thrive

1. **Make U.S. government datasets available to MI developers and users.**
2. **Work with state and local governments and foreign governments to open up datasets to public use.**
3. **Establish standards and taxonomies that ensure that data sets made available through open-data initiatives are machine readable and interoperable.**
4. **Encourage private-sector data sharing and novel uses of data while protecting privacy and data security.**
5. **Defend the free flow of data across national borders and combat data localization.**

D.

Create Thoughtful Public Policy to Spur MI Adoption and Reduce Barriers

STRATEGY
D

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Policymakers should also pursue measures that help spur demand for MI technologies, growing the market for MI tools and giving room for new U.S. MI enterprises to expand. Currently, enterprise adoption of MI is low. A September 2017 study released by the MIT Sloan Management Review found that only 5 percent of surveyed companies had extensively incorporated MI into their offerings, and less than 40 percent had any MI strategy in place. A range of barriers exist for companies interested in adopting MI tools, including a lack of IT infrastructure, lack of access to talent, uncertainties over regulations and customer rights, and high cost of implementation. While many of these issues can be addressed at the enterprise lev-

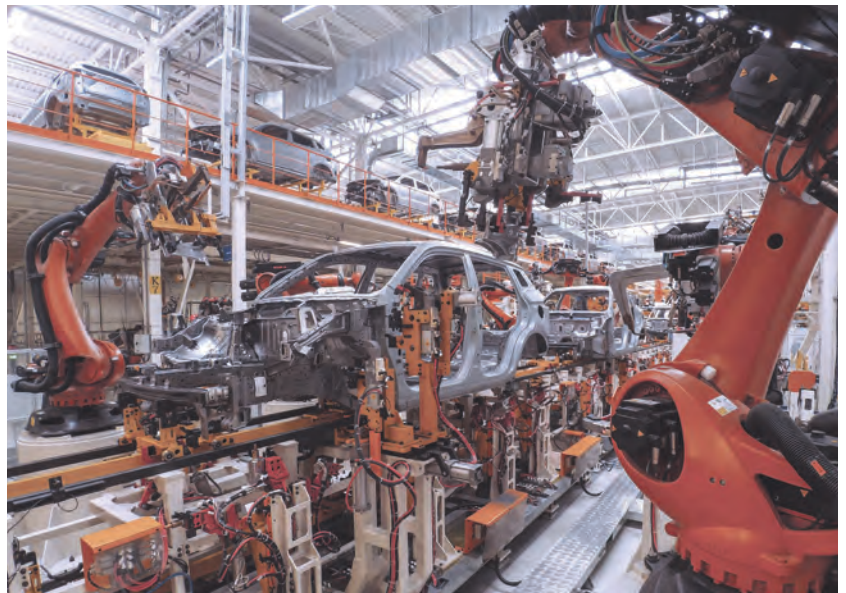
el, the federal government can support the adoption of automation and MI by firms by removing legal and regulatory obstacles and creating incentives for MI adoption.

For businesses, lack of adequate IT infrastructure capable of supporting MI is a key barrier to deploying the technology. A survey conducted by Vanson Bourne and Teradata found that 40 percent of respondents cited lack of IT infrastructure as a key barrier to MI realization. MI adoption is consistently higher among organizations that have already invested in the necessary supporting infrastructure, including big data and cloud computing. To support the adoption of IT solutions that will

"For businesses, lack of IT infrastructure capable of supporting MI is a key barrier to deploying the technology."

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enable future MI deployment, the U.S. government should look into partnering with lenders and technology companies to create loan guarantee programs to support IT modernization. The program could fund adoption of cloud solutions and data analytics, partnering borrowers with private-sector experts to help companies integrate these new services into their operations and prepare for wider adoption of MI tools. Tax incentives like the research and experimentation (R&E) tax credit could also be expanded to encourage businesses to put resources towards developing their own internal machine intelligence services.



Federal procurement policies may also be leveraged to provide guidance to the private sector on the responsible adoption of MI technologies. In the defense sector, this will mean reforming the infamously convoluted procurement process to enable quicker adoption of new MI technologies. DoD efforts like the Silicon Valley-based DIUx should be encouraged as opportunities to create new and streamlined pathways toward the acquisition of new advanced technologies. Throughout the rest of

Automated
production
line at
automobile
plant.

"Many businesses face barriers to deployment due to uncertainty over the application of existing legal and regulatory regimes to the case of machine intelligence."

**STRATEGY
D**

the federal government, it will be essential for federal IT modernization efforts to promote the adoption of MI and enabling technologies within the U.S. government.

► **THE** U.S. government can also work to eliminate regulatory uncertainty and questions around liability that arise from MI. Many businesses face barriers to deployment due to uncertainty over the application of existing legal and regulatory regimes to the case of machine intelligence. An example of this is in the automated vehicles (AV) industry, where Congress has recently been working to update the current federal safety standards to accommodate self-driving cars and prevent the emergence of a patchwork of state regulations governing the development and deployment of AVs.

As a first step, the president should instruct all federal regulatory agencies to initiate a voluntary consultation process with companies on potential applications of MI in their industries and where regulatory uncertainty is impeding the adoption of MI. The goal of these consultations should be to identify steps that can be taken to reduce regulatory uncertainty and promote the safe and responsible adoption of MI across the sector.

One critical area to clarify is in the domain of liability. As MI-powered products like self-driving cars or medical diagnostic tools become more widespread, mistakes and accidents caused by these systems will become more common. Especially in situations where a human and an MI system are working in tandem, the extent of each party's relative liability will be a thorny ques-

tion that courts must resolve before enterprises can assess their potential risk when developing or adopting MI tools.

There are two initial steps that can be taken to address legal uncertainty around the use of MI. First, the government should commission a panel of legal and technical experts to examine existing law and precedent and identify areas where the use of MI creates questions around the application of core principles of U.S. law. This panel should be given a mandate to articulate a set of legal principles that can be used in the governance and application of liability to MI systems that act autonomously or in concert with humans, and how these principles can be implemented through legislation or through interpretation of existing law.

Second, the findings of this panel should inform a training program for judges and attorneys to provide basic technical literacy around existing MI technologies and an overview of the application of legal principles to MI applications. Judicial interpretation and precedent will play a key role in defining the legal landscape for MI. When the Human Genome Project transformed forensics and criminal justice, the Department of Justice, Department of Energy, and National Institutes of Health invested in education programs for judges and attorneys to ensure that the legal system was prepared for new uses of DNA evidence. The implications of machine intelligence for the legal system will likely be far broader and even more complex, and our justice system must respond with appropriate urgency to ensure it has the technical capacity to handle the questions that will begin to emerge. ▲

RECOMMENDATIONS

Create thoughtful public policy to spur MI adoption and reduce barriers

1. Incentivize firms to invest in IT modernization to support MI.
2. Instruct regulators to initiate voluntary consultations with the private sector on key steps to reduce regulatory uncertainty and support safe and responsible adoption of MI across their sectors.
3. Establish a commission of legal and technical experts to examine the application of core principles of U.S. law to MI.
4. Develop a training program for judges and attorneys on current MI technologies and the application of legal principles to MI.

E.

Develop Proactive Strategies to Manage MI's Risks

STRATEGY
E

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MI technology continues to evolve, and will likely look very different in a few years' time, but there are many risks and challenges around the technology that must be proactively addressed. A well-thought-out strategy to manage MI's challenges can help build consumer trust in the technology and help us to avoid falling into a trap of reactionary policymaking when accidents inevitably happen. At the same time, we must ensure that any rules and controls we put in place make sense for the technology and do not strangle the commercial MI industry.

The first, and perhaps most difficult, risk to address will be bias and discrimination. MI systems are designed to

shape our behavior and perform tasks on our behalf, meaning that when they demonstrate bias they can have profound impacts on human opportunity. MI systems often exhibit implicit bias, either imparted inadvertently by their creators or by the data on which they were trained. Google's ad targeting software, for example, has been found to target job ads for high-paying executive positions far more frequently to male web users than female ones. This discrimination was not intentional, however, with researchers instead suggesting that Google had simply "lost control over its massive, automated advertising system," noting that "Even without advertisers placing inappropri-



Waymo CEO John Krafcik delivers a speech about self-driving cars at the 2017 Web Summit in Lisbon on Nov. 7, 2017.

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PATRICIA DE MELO
MOREIRA/AFP/
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ate bids, large-scale machine learning can behave in unexpected ways.”

It will also be crucial to manage the risks of physical harm that come as MI systems come to play an increasing role in controlling real-world systems. The most salient example of safety risks is autonomous vehicles, where failure of sensors or other equipment can lead to fatal consequences. However, AVs are only the first wave of what promises to become a flood of new robotic systems interacting with us on a daily basis. The coming ubiquity of drones in our skies and robots in our homes will demand the active management of safety risks, not only to protect against isolated malfunctions, but also against cascading failures of networked systems.

Perhaps the most visible and widely discussed risk associated with machine intelligence is the issue of control. While concerns over the emergence of malevolent MI systems are badly overblown, this should not distract us from the importance of developing robust incentive structures and ethical

systems to guarantee that MI systems operate with our best interests in mind. For example, now that MI programs can lie, it will be up to humans to ensure that they do not. The United States had its first taste of the damage that could be done by autonomous systems in 2010, when automated stock-trading programs caused a “flash crash” that erased nearly \$1 trillion of wealth in minutes. As we prepare to turn an increasing share of the management of our society over to MI systems, it will be of paramount importance that we ensure these programs are both predictable and controllable by humans to prevent such destructive failures from occurring in the future.

The government will have to play a major role in managing the risks of MI technologies. While apocalyptic warnings by some in the tech community are overblown, we will not be able to ignore the range of more subtle risks posed by issues of privacy, algorithmic bias, system safety and control. As individuals increasingly come to rely on MI technologies to manage social services and support decisionmaking in sensitive domains like health care,

employment, and criminal justice, the government will have a responsibility to ensure that these technologies are being deployed appropriately. It is unrealistic to expect that MI systems will ever be perfectly error-free, so government should instead focus on managing the risks that we can anticipate and establishing clear plans and expectations to deal with failures when they do arise.

To help policymakers understand and appropriately respond to emerging MI risks, the federal government should establish an advisory committee of experts from industry and academia to support government decisionmaking about MI, similar to the proposed FUTURE of AI Act. The committee could provide technical expertise and expert advice to the Commerce Department as they consider the implications of MI in the United States, addressing not only risk management but also workforce development, research priorities, data accessibility, and ethical standards. The committee would help the department to identify barriers to innovation and growth, and provide recommendations on how to address emerging issues in MI in a way that avoids stifling innovation.

The U.S. government should create a new initiative through the National Institute of Standards and Technology (NIST) to develop safety, ethics, and control standards for MI applications. NIST, in cooperation with industry and international partners, should articulate a set of technical and governance standards for safe and responsible MI development and use. The U.S. government should also step up engagement with international standards-making bodies to promote global standards for safe and ethical MI.

NIST should also initiate voluntary consultations with the private sector on the development of a testing and certification regime for MI systems. Contextual audits and other testing mechanisms may be the only way to provide oversight with MI systems too complex to understand through source code analysis. Federal agencies should be directed to engage with industry partners and NIST to examine the viability of using testing to manage safety, control, and discrimination risks

for programs adopted by the federal government in sensitive or high-risk domains.

Even with clear standards for MI development and use and a testing regime to anticipate and address problems, it is unreasonable to expect that MI systems will ever be perfectly predictable or error-free. Mistakes will inevitably be made by MI programs, and accidents will occur. Federal regulators should be proactive in articulating a set of expectations for how MI developers should respond to system failures. Clarifying expectations will not only reduce uncertainty for MI companies, it will help guide efforts to build control mechanisms into MI systems and assist in heading off public uproar in the event of MI accidents.

The advance of MI will also pose new privacy risks for individuals as data-collection activities are expanded to power new MI systems. As more of our devices become connected to the internet, the granularity of data available to corporations, marketers, and governments about our daily lives will introduce new possibilities for scoring and surveillance. And even if the companies collecting our data commit to its responsible use, consumers still face risks from outside parties capable of stealing sensitive information or combining anonymized data in ways that allows for re-identification of users.

While the rise of big data has led to the normalization of increasingly detailed data collection from technology users, consumer trends have clearly indicated that most citizens value personalized services more than maintaining strict control of the data they produce. As such, it would be unwise for the United States to consider implementing a strict data privacy regime such as the European Union's General Data Protection Regulation (GDPR) at the risk of stifling innovation. At the end of the day, however, the United States will not be the only market American technology companies will target. The government should thus work with industry to ensure that MI systems developed in the United States remain internationally viable in the face of advancing data regulations in places like the European Union.

Though the adoption of a GDPR-like data privacy regime would be unwise in the context of ensuring continued American innovation, the absence of regulation will only make it more important for the government to actively engage with industry through other channels to proactively manage the emerging privacy risks associated with big data and machine intelligence. In particular, the widespread availability of user data collected through online services and connected devices will create new incentives for companies to share data. The use of data for purposes outside the context of its initial collection threatens user control and privacy, and risks eroding consumer trust throughout the digital economy. In response, the government should work with a transparent, industry-led process to develop standards and codes of conduct for data collection and use to manage the privacy risks created by MI.

The United States should be careful, however, to avoid overburdening the MI industry with restrictive regulations. In countries like China, MI developers have an advantage over their American

and European counterparts due to a lack of regulatory obstacles, especially in the crucial areas of real-world trials and mass data collection. As former Google China president Kai-Fu Lee states, “in driverless cars, the United States has about a two-year head start, but each minute American tech companies find themselves mired in a regulatory battle or hobbled by objections from transportation incumbents, the Chinese can close that gap.” Regulation also damages Europe’s competitiveness. Stringent EU regulations on the collection and storage of personal data greatly complicates algorithmic development, and their restrictions on automated decisionmaking make the EU market hostile to MI companies. The United States should be careful not to share this fate, and should be wary when considering protections for user privacy to allow sufficient flexibility to maintain the innovative capacity of U.S. businesses while simultaneously managing privacy and safety risks for individuals. ▲

RECOMMENDATIONS

Develop proactive strategies to manage MI's risks

1. Establish a U.S. MI Advisory Committee through legislation like the FUTURE of AI Act to advise the Commerce Department on MI governance.
2. Develop technical and governance standards for safe and responsible MI.
3. Articulate clear expectations for companies that develop and deploy MI in the event of accidents.
4. Consult with industry on the feasibility of a testing regime for MI.
5. Engage with industry leaders to devise an internationally viable privacy regime for the MI era.

F.

Lead the Development of MI Technology and MI Governance around the World through Strategic MI Partnerships

STRATEGY
F

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Leadership does not mean building our MI capabilities alone. International partnerships can help us to develop and deploy technology more quickly, and share our values, approaches, and standards with the rest of the world.

First, we should establish research partnerships with other MI leaders to develop new MI technologies. Countries like Canada and the United Kingdom have been historic leaders in the field of MI due to the strength of the research from academics and startups in cities like London, Toronto, and Montreal. Strengthening the United States' ability

to collaborate with technology leaders in these countries will significantly improve our capacity for innovation and help keep the United States on the cutting edge of new MI development. In particular, policymakers can support the creation of exchange programs for MI researchers and students, ensure an open and streamlined immigration process for experts from these nations, and create new opportunities for investment to flow between our countries. In the case of Canada, the United States could also benefit from the establishment of “innovation corridors” like the one proposed between Seattle

and Vancouver, which could improve access to talent and investment for MI developers on both sides of the border.

We should also look to leverage the comparative advantages of other countries whose strengths complement our own. The United States has established clear global leadership in the development of machine cognition and other MI software domains, but in areas like smart manufacturing and robotics, our country still lags places like Germany and Japan. Policymakers should look to build partnerships that would allow our countries to build off each other's capabilities. An example of this is the

work done by Germany's Industrie 4.0 and the U.S. Industrial Internet Consortium to develop common standards for intelligent manufacturing that leverage both the United States' strength in software and big data, and Germany's manufacturing expertise.

As MI extends the reach of digital systems into the physical world, it will place a significant burden on natural resources, particularly rare earth elements used in computing and batteries and metals and plastics used in constructing robots and cyber-physical systems. The growth of MI and associated technologies will likely continue



"Strengthening the United States' ability to collaborate with technology leaders in these countries will significantly improve our capacity for innovation and help keep the United States on the cutting edge of new MI development."

to fuel demand for these raw materials, many of which are produced in developing regions, for example South America's "lithium triangle," which produces more than half of the world's lithium. China is making significant investments in companies and infrastructure and developing extensive political partnerships with these countries in order to secure control of their natural resources, recognizing the importance of these raw materials to Chinese competitiveness in emerging industries like autonomous and electric vehicles. The United States should do the same.

Finally, the United States should step up its participation in international MI standards and governance bodies to strengthen its influence over the future direction of artificial intelligence. Organizations like the International Telecommunication Union (ITU), International Organization for Standardization (ISO), and Institute of Electrical and Electronics Engineers (IEEE) are al-

ready beginning to establish initiatives to develop technical, ethical, and governance standards for the development of MI, and the decisions reached by those involved will play a powerful role in shaping the future of the technology.

China is already racing to gain influence over these bodies, with its recent Next Generation Artificial Intelligence Development Plan including explicit calls for the country's MI enterprises to send representatives to these meetings and help take the lead in formulating new international standards. The United States should be assertive in promoting its own vision for MI in these organizations, but first the United States must articulate what that vision should be. An MI national strategy would give policymakers the opportunity to establish a clear set of values the United States believes should govern the future of MI, and enable our country to more productively engage with the international community in shaping MI's future. ▲

RECOMMENDATIONS

Lead the development of MI technology and MI governance around the world through strategic MI partnerships

1. Establish research partnerships with allied nations to develop new MI technologies.
2. Leverage the comparative advantages of allies in MI-enabling technologies.
3. Establish resource partnerships with developing nations.
4. Step up participation in international technical and governance standards bodies to shape the global trajectory of MI technology.

CONCLUSION

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It will take time to craft and implement all of the elements of a national MI policy, but we must act now to establish a cohesive national MI strategy. As a first step, the president should designate an MI lead at the White House to coordinate the development and implementation of a strategy and ensure that this is a priority across the federal government. This person should prioritize reviewing and implementing the 2016 AI R&D Strategy recommended by the National Science and Technology Council. Federal agencies and regulators should also initiate consultations with the private sector and MI experts in academia to ensure that our approach is flexible enough to adapt over time and appropriate to the technology. These discussions will take time, but will help to guide the efforts of policymakers and ensure that MI governance develops in partnership with industry, not in opposition to it.

MI could usher in a new era of growth and prosperity in the United States, fueling significant gains in growth and

productivity and new opportunities throughout our society. Currently, MI in the United States is advancing in a policy vacuum. The federal government lacks a coordinated plan for responding to the emergence of MI, risking the emergence of a patchwork of state and local regulations that could impede innovation. Other countries have already established national strategies to capture the benefits of MI, and are aggressively pursuing their implementation as they race to catch up with the United States. Policymakers must clearly articulate the values by which MI should be governed, and work to incorporate those values into the design of MI systems developed within the United States. At the same time, we need a strategy to ensure that the benefits of MI accrue to all, and that the risks from MI are responsibly managed. If we wait for a major accident to provoke action we are likely to get the kind of disruptive and reactive policymaking that stifles innovation and undermines U.S. leadership. ▲

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